

ULC405 : PRINCIPLES OF POWER SYSTEM ENGINEERING

L	T	P	Cr
3	1	0	3.5

Course Objectives: The course aims to impart a critical theoretical and practical introduction to electrical, mechanical design of transmission system and steady state analysis of network.

Syllabus

Introduction to Power System: Structure of power systems, Growth of power systems-Indian overview, Interconnections and their advantages. Power Supply Systems and their comparison.

Electrical Design of Transmission Line: Choice of voltage and frequency, Types of conductor, Size of conductor, Resistance, Inductance and capacitance of single phase and three phase transmission lines. Effect of ground on Capacitance. Parameters of Insulated Cables.

Mechanical Design of Transmission line: Tension and sag calculations, Factors affecting Sag, Sag template, Stringing charts, Vibrations and vibration damper.

Insulators: Insulator types, String efficiency and its Improvement.

Performance of Transmission Lines: Characteristics and performance of power transmission lines: Short, Medium, Long lines, Generalized constants, Power flow, regulation, Series and shunt compensation, Corona, Ferranti Effect, Electrostatic and Electromagnetic interference with communication lines.

Transmission network Calculations: Single line diagram of power system, Per Unit System and its advantages, Admittance model, Bus admittance matrix assemble.

Load Flow Studies: Load flow problem, Power flow equations, Load flow solution using Gauss Seidal and Newton Raphson methods, Comparison of load flow methods.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. **Evaluate the electrical parameters of various configurations of transmission lines and insulated cables**
2. **Carry out** the mechanical design (sag and tension) of transmission line under various environment and geographical conditions
3. **Analyse the transmission line models to evaluate its performance**
4. Carry out load flow analysis of **power system network**

Text Books:

1. Nagrath, I.J. and Kothari, D.P., Power System Engineering, Tata McGraw Hill (2007).
2. Stevenson, W.D., Power System Analysis, McGraw Hill (2007).
3. Gupta, B.R., Power System Analysis and Design, S. Chand and Company Limited (2009)

Reference Books:

1. M. Pai, Computer Techniques in Power System analysis, Tata McGraw Hill, 2nd ed., (2005).
2. Chakraborti, A., Soni, M.L., Gupta, P.V. and Bhatnagar, U.S., A Text Book on Power System Engineering, Dhanpat Rai and Co. (P) Ltd. (2008).
3. Elgard, O.L. , Electric Energy Systems Theory, McGraw Hill Publications , 2nd ed., (2017).

Evaluation Scheme

Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	70
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	30